Harshvardhan Kumbharkar

B.Tech Artificial Intelligence & Data Science Graduate | Backend & AI Developer
AISSMS Institute of Information Technology, Pune

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Professional Summary

Artificial Intelligence and Data Science graduate with hands-on experience developing Python-based backend systems, REST APIs, machine learning applications, and computer vision solutions. Built AI systems using FastAPI, YOLOv8, InsightFace, NLP, and cloud deployment platforms. Published two research papers covering AI applications in video games and construction-site safety monitoring.

Education

AISSMS Institute of Information Technology, Pune
B.Tech in Artificial Intelligence and Data Science | Aug 2022 – Jun 2026
CGPA: **7.19 / 10**

Airforce College, Viman Nagar, Pune
Higher Secondary Education | 2020 – 2022

Bishop's Co-Ed School, Undri, Pune
CISCE | 2020

Technical Skills

- **Programming:** Python, C++, SQL
- **Backend Development:** FastAPI, REST APIs, WebSockets, Backend Architecture, Database Design
- **AI & Machine Learning:** Machine Learning, Deep Learning, NLP, Computer Vision, Feature Engineering
- **AI Frameworks & Libraries:** PyTorch, YOLOv8, InsightFace, OpenCV, NumPy, Pandas, Scikit-learn
- **Cloud & Deployment:** AWS Fundamentals, Render
- **Data & Visualization:** Power BI, Tableau

Projects

NyayaSathi – AI Legal Study Assistant (Local LLM) | GitHub

- Built a free, fully local AI assistant for Indian law students that turns plain-language queries (e.g. "cheque bounce", "land acquisition") into the relevant **statute sections** and **leading Supreme Court cases** with real reporter citations.
- Developed a retrieval pipeline over Indian Kanoon that parses **10 major Acts (3,300+ sections)** and automates old-to-new criminal law mapping (IPC/CrPC/IEA → BNS/BNSS/BSA) down to sub-section level, e.g. IPC 420 → BNS 318(4).
- Grounded **Qwen3 8B (via Ollama)** answers in the retrieved provision text with enforced source citations, and kept case data deterministic to prevent hallucinated holdings.
- Built a **citation verifier** that normalises SCC/AIR/SCR formats and flags wrong, mismatched or fabricated case citations; **FastAPI** backend with rate-limited, robots.txt-compliant fetching.

AI-Based Helmet Monitoring & Face Recognition System | GitHub

- Developed a real-time computer vision system combining **YOLOv8 helmet detection** with **InsightFace facial recognition** to identify helmet violations and associate detections with worker identities.

- Implemented bounding-box spatial verification to determine helmet compliance and used normalized facial embeddings with cosine similarity for identity matching.
- Achieved a reported **96.1% mAP@0.5** and **69.2% mAP@0.5:0.95** for the YOLOv8 detection model.
- Built the backend using **FastAPI and WebSockets** for asynchronous real-time frame processing and communication with the frontend.

Public Service Management System – Deep Learning Deployment | GitHub

- Developed an AI-powered system to classify and prioritize citizen service requests using structured and unstructured datasets.
- Implemented deep-learning-based categorization to automate request processing and reduce reliance on manual classification.
- Designed the application for cloud accessibility and deployed the system using **Render**.
- **Result:** Automated the request categorization workflow and reduced manual processing requirements.
- **Scale / Performance:** [ADD VERIFIED NUMBER — e.g. dataset size, classification accuracy, processing time, or number of requests tested]

AI-Based Chatbot – NLP Application | GitHub

- Developed a Python-based conversational chatbot using NLP techniques for text preprocessing, intent classification, and response generation.
- Implemented tokenization, stemming, vectorization, and intent-based classification to map user messages to predefined conversational categories.
- Designed structured response logic to improve consistency across supported conversation flows.
- **Result:** [ADD VERIFIED RESULT — e.g. intent classification accuracy, number of intents, number of training samples, or response latency]

Resume Management System

- Designed a backend-oriented application for storing and managing structured user resume profiles using database-driven architecture.
- Implemented API integration concepts and structured data handling to support profile creation, retrieval, and organization.
- Focused on modular backend architecture and database design for maintainability and future feature expansion.
- **Result:** [ADD VERIFIED RESULT — e.g. number of API endpoints, database records tested, response time, or users supported]

Metadata Creation & Dataset Structuring

- Generated and organized metadata for **7 structured and unstructured datasets** to improve dataset discoverability and usability.
- Applied data cleaning, normalization, categorization, and documentation techniques to create consistent metadata structures.
- **Result:** Produced standardized dataset documentation and metadata structures for improved data organization and downstream analysis.

Research Publications

Emergent Minds: The Evolution and Future of AI in Video Game NPCs
Published in IJFMR. Research analyzing the progression from rule-based NPC systems toward adaptive AI-driven behavioral models for dynamic game interactions.
DOI: https://doi.org/10.36948/ijfmr.2025.v07i03.42387

AI-Based Helmet Monitoring and Face Recognition System at Building Construction Site: A Survey
Survey research analyzing computer vision and facial recognition approaches for automated construction-site safety monitoring and compliance.
DOI: https://doi.org/10.48175/IJARSCT-29820

Certifications

- **Microsoft Azure AI Fundamentals (AI-900)** — Certification Number: 9BD3E0-E8839B
- **Oracle Academy – Java Foundations**
- **Oracle Academy – Database Programming using SQL**
- **UiPath – RPA Developer Foundation Certificate**
- **AWS Academy Graduate – AWS Academy Cloud Foundations** — 20 Hours | Issued Jul 2025

Experience

Internshala Campus Ambassador | 2023 | 3 months

- Led campus outreach initiatives promoting internship and skill-development programs through direct student engagement.
- Coordinated peer communication and promotional activities to increase awareness of available internship and learning opportunities.
- Developed practical experience in communication, outreach, leadership, and stakeholder engagement.

Languages

- English
- Hindi
- Marathi
- Japanese – JLPT N5 Certified